

S. M. Mohiuddin Khan Shiam

📍 Akbar Shah Railway Housing Society, Chattogram, Bangladesh

✉️ mohiuddin.khan.shiam@gmail.com | 📧 mohiuddin.khan.shiam@g.bracu.ac.bd



CAREER OBJECTIVE

Motivated Computer Science graduate with strong academic grounding in software systems, artificial intelligence, and peer-reviewed machine learning research. Seeking graduate-level Computer Science and software/AI engineering opportunities to apply solid algorithmic problem-solving, rigorous analytical thinking, and practical software development skills, with a long-term goal of pursuing a Ph.D. abroad in advanced artificial intelligence.

RESEARCH INTERESTS

Machine Learning, Deep Learning, Computer Vision, Explainable AI (XAI), Time Series Forecasting, Quantum Machine Learning (QML), and Medical Image Diagnostics.

ONLINE PROFILES & RESEARCH IDENTIFIERS

Academic & Scientific Identifiers:

- 🎓 **Google Scholar:** [Mohiuddin Khan Shiam](#)
- 🆔 **ORCID iD:** [0009-0005-5504-2595](#)
- 📖 **Scopus Author ID:** [60712607800](#)
- 🔬 **ResearchGate:** [S-M-Mohiuddin-Khan-Shiam](#)
- 📄 **SSRN Author Profile:** [SSRN: 12781549](#)
- 📊 **SciProfiles:** [mohiuddin-khan-shiam](#)
- 🎓 **Academia.edu:** [bracu.academia.edu/shiam](#)
- 📊 **Figshare:** [S_M_Mohiuddin_Khan_Shiam](#)

Development & Open-Source Platforms:

- 🌐 **Portfolio Website:** [mohiuddin-khan-shiam.github.io](#)
- 🐙 **GitHub Profile:** [mohiuddin-khan-shiam](#)
- 🌐 **LinkedIn Profile:** [s-m-mohiuddin-khan-shiam](#)
- 🏆 **Kaggle Profile:** [simmohiuddinkhanshiam](#)
- 👤 **Hugging Face:** [mohiuddin-khan-shiam](#)
- 🧬 **Rosalind (Bioinformatics):** [shiam](#)

EDUCATION

BRAC University

Bachelor of Science in Computer Science

Dhaka, Bangladesh

2026

Hazera-Taju Degree College

Higher Secondary Certificate (HSC) – Science Group

Chattogram, Bangladesh

2020

Nasirabad Govt. High School

Secondary School Certificate (SSC) – Science Group

Chattogram, Bangladesh

2018

ACADEMIC THESIS

Title: Deep Learning Frameworks for Suspicious Activity Detection

2026

Supervisors: Prof. Dr. Amitabha Chakrabarty & Prof. Dr. Md. Golam Rabiul Alam | **Code:** 🐙 [GitHub: Nano3D](#)

- Benchmarked deep spatiotemporal learning architectures under uniform conditions across 5 public surveillance datasets: UCF-Crime, XD-Violence, RWF-2000, UBI-Fights, and SCVD.
- Proposed **Nano3D**, an ultra-compact 3D convolutional model containing only 486K parameters (1.86 MB model size), achieving 88.7% accuracy, 0.947 AUC, and 7.19 ms inference latency.
- Integrated Grad-CAM spatiotemporal saliency heatmaps to visually explain and attribute automated anomaly localization.

RESEARCH EXPERIENCE

A. Published Articles

HybridStack: Explainable hybrid ensemble for accurate inflation forecasting with high-frequency macro-financial data 2026
Array, Elsevier | **Rank:** Q1 | **Impact Factor:** 5.3 | **CiteScore:** 10.1
Authors: S. M. Mohiuddin Khan Shiam, Meherunnesa Neela, Amrijit Biswas, and Mahdy Rahman Chowdhury
Citation & Link: Vol. 31, Art. 101003, ISSN: 2590-0056 | DOI: [10.1016/j.array.2026.101003](https://doi.org/10.1016/j.array.2026.101003) | ScienceDirect: [Article Link](#)

- Engineered a multi-tier hybrid ensemble combining gradient-boosted trees (XGBoost, LightGBM, CatBoost) with regularized meta-regressors.
- Optimized hyperparameters via Bayesian search across 10 macro-financial sources, yielding a held-out test RMSE of 0.0006.
- Integrated SHAP, LIME, and Partial Dependence Plots (PDP) to generate policy-grade interpretability of economic predictors.

B. Manuscripts Submitted / Under Review

Nano3D: An Explainable Ultra-Lightweight Deep Learning Model for Surveillance Video Anomaly Detection 2026
Target Journal: Scientific Reports (Nature Portfolio) | **Rank:** Q1 | **Impact Factor:** 4.9
Authors: S. M. Mohiuddin Khan Shiam, et al. | **Code Repository:** [GitHub: Nano3D](#)

- Formulated a resource-efficient 3D-CNN architecture (486K parameters, 1.86 MB footprint) for edge-device video anomaly detection.
- Attained 88.7% accuracy and 0.947 AUC with 7.19 ms latency across 5 canonical benchmark datasets, outperforming heavier baselines.
- Utilized Grad-CAM spatiotemporal heatmaps for interpretable verification and localization of suspicious surveillance sequences.

C. Ongoing Research

- Quantum Machine Learning (QML):** Formulating hybrid quantum-classical algorithms, Parameterized Quantum Circuits (PQCs), and Variational Quantum Classifiers (VQC) using Qiskit and PennyLane for high-dimensional constrained optimization.
- AI in Medical Image Diagnostics:** Designing 3D Vision Transformers (ViT) and CNNs for early cancer detection (mammography, histopathology) and acute ischemic stroke lesion segmentation from brain CT/MRI scans with Grad-CAM and SHAP.

WORK & LEADERSHIP EXPERIENCE

Machine Learning Researcher 2026
Mahdy Research Academy Dhaka, Bangladesh

- Conducted advanced AI research in high-frequency time-series forecasting, neural networks, and Bayesian-tuned hybrid ensemble models.
- Engineered end-to-end ML data pipelines and authored peer-reviewed scientific articles published in international Q1 journals.

Treasurer 2024
IEEE BRAC University Student Branch Dhaka, Bangladesh

- Managed student branch fiscal budgeting, financial tracking, funding allocations, and transparent institutional reporting.
- Co-organized major technical workshops, seminars, and training programs, including the flagship “Smart City & IoT” workshop.

TECHNICAL PROJECTS

Comparing Deep Learning and Statistical Models for Stock Price Prediction 2024
Repository: [GitHub: Stock-Price-Prediction](#)

- Conducted empirical time-series forecasting evaluating LSTM recurrent neural networks against classical ARIMA statistical models.
- Forecasted HP Inc. historical equity price trends, analyzing stationarity, volatility dynamics, and model trade-offs.

SkillShikhi: Collaborative Skill-Sharing & Learning Web Platform 2024
Repository: [GitHub: SkillShikhi](#)

- Built a full-stack educational and skill-sharing web platform utilizing Next.js, React, Node.js, and MongoDB with MVC architecture.
- Implemented JWT-based authentication, user dashboards, dynamic content routing, responsive UI, and secure RESTful APIs.

Autonomous Line Following Hardware Robot 2023

- Built an autonomous line-following vehicle for the Digital Logic Design curriculum using discrete logic gates, IR optical sensors, comparator ICs, and H-bridge motor drivers without microcontrollers, mastering low-level digital logic design.

OPEN SOURCE CONTRIBUTIONS

Active contributor to 11+ global open-source organizations across 17+ mission-critical repositories powering millions of developers.

- **Microsoft Ecosystem:** [VS Code](#) (IntelliSense & debugging API), [Windows Terminal](#) (performance & shell integration), [WSL](#) (cross-platform filesystem), [PowerToys](#) (system productivity utilities), [AutoGen](#) (multi-agent AI orchestration), [vcpkg](#) (package management), [WinGet CLI](#) (command-line package discovery).
- **Artificial Intelligence & Machine Learning:** [scikit-learn](#) (core ML algorithm optimization & estimators), [Keras](#) (model architecture layers & training pipelines), [Ultralytics YOLO](#) (real-time computer vision tuning), [OpenVINO](#) (neural network inference acceleration).
- **Cloud Infrastructure & DevOps:** [Kubernetes](#) (controller stability & scheduler optimizations), [Kops](#) (automated cluster provisioning), [Apache Airflow](#) (DAG workflow scheduling & operators), [Flux CD](#) (GitOps sync & Helm integration).
- **Scientific Research & Education:** [The Algorithms \(Python\)](#) (clean algorithm implementations), [Manim](#) (mathematical animation engine), [Boltz](#) (biomolecular interaction modeling for structural biology).

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, MATLAB, JavaScript, TypeScript, SQL, HTML5/CSS3, PHP, Bash/Shell.
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, OpenCV, Pandas, NumPy, SciPy, Matplotlib, Seaborn.
- **AI Paradigms & Methods:** 3D CNNs, ConvLSTM, Vision Transformers (ViT, ViViT), Explainable AI (SHAP, LIME, Grad-CAM), Hybrid Ensembles, Bayesian Optimization, Time-Series Forecasting, Quantum Machine Learning (Qiskit, PennyLane).
- **Web & Cloud Technologies:** Next.js, React, Node.js, Express, MongoDB, RESTful APIs, Docker, Kubernetes, Linux/Unix, Git, GitHub, Overleaf, $\LaTeX$, VS Code.

AWARD AND ACHIEVEMENT

- **Top 10% Global Talent** | Issued by *The Talent Games*, Nov. 2024.
- **RS-60 Cricket Cup Champion** | Issued by *BRAC University*, Jul. 2022.
- **Academic Excellence in National Board Examinations:** Awarded perfect GPA 5.00 / 5.00 in the Higher Secondary Certificate (HSC 2020), Science Group.

CERTIFICATIONS

- | | |
|--|------------------|
| • Enhance Your Leadership Skills LinkedIn Learning | Issued Jan. 2025 |
| • Agile Foundations International Institute of Business Analysis™ (IIBA®) | Issued Dec. 2024 |
| • Time Management Fundamentals Project Management Institute (PMI) | Issued Aug. 2024 |
| • Communication Hacks 10 Minute School | Issued Jun. 2024 |

PROFESSIONAL REFERENCES

1. Dr. Md. Golam Rabiul Alam

Professor, Department of Computer Science & Engineering
School of Data & Sciences, BRAC University
✉ rabiul.alam@bracu.ac.bd | [Faculty Profile](#)
🎓 [Google Scholar](#) | [ORCID](#) | [Scopus](#)
🔬 [ResearchGate](#) | [IEEE](#) | [LinkedIn](#)

3. Dr. Amitabha Chakrabarty

Professor, Department of Computer Science & Engineering
School of Data & Sciences, BRAC University
✉ amitabha@bracu.ac.bd | [Faculty Profile](#)
🎓 [Google Scholar](#) | [ORCID](#) | [Scopus](#)
🔬 [ResearchGate](#) | [LinkedIn](#)

2. Dr. Mahdy Rahman Chowdhury

Professor, Department of Electrical & Computer Engineering
North South University
✉ mahdy.chowdhury@northsouth.edu | [Faculty Profile](#)
🎓 [Google Scholar](#) | [ORCID](#) | [Scopus](#)
🔬 [ResearchGate](#) | [LinkedIn](#)

4. Md. Ariful Islam Sanim

Ph.D. Student & Graduate Teaching Assistant in Statistics
University of South Carolina, USA
✉ msanim@email.sc.edu | [Department Directory](#)
🎓 [Google Scholar](#) | [ResearchGate](#)
[LinkedIn](#): [md-ariful-islam-sanim](#)

DECLARATION

I hereby declare that the above information given in this CV is true to the best of my knowledge.

Place: Akbar Shah Railway Housing Society, Chattogram
Date: September 2026

S. M. Mohiuddin Khan Shiam
Department of Computer Science & Engineering
BRAC University, Dhaka, Bangladesh